



UNIVERSITY MALAYSIA TERENGGANU
FACULTY OF COMPUTER SCIENCE & MATHEMATICS

SOFTWARE ARCHITECTURE
CSE3433

INDIVIDUAL LOGBOOK

GROUP NO. 16

SERVICE-ORIENTED SMART CAMPUS SUPPORT SYSTEM

PREPARED BY:
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PREPARED FOR:
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[BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING)
WITH HONOURS]
SEMESTER 2 2025/2026

1.0 Team Information

Name	Matric No	Role
MUHAMMAD UQAIL RAFIQIE BIN MOHD FARIS	S76227	DATABASE ADMIN

2.0 Weekly Progress Report

Week 4

Item	Description
Week	Week 4 (2 - 8 March 2026)
Weekly Activities	During this week, the team discussed the project idea and analysed the main problems that the Disaster Relief Coordination Platform aimed to solve. The team identified different users and the information that needed to be stored by the system.
Individual Contributions	As the Database Administrator, I contributed by identifying the main data entities required for the system, including users, disaster reports, volunteer tasks, resources and notifications. I also discussed with the team how information should be stored and connected between different modules.
Challenges	The main challenge was identifying the correct database structure because the system involved multiple user roles and different types of information that needed to interact with each other.
Action Plans	Prepare an initial database structure and identify possible table relationships before implementation begins.

Week 5

Item	Description
Week	Week 5 (9 - 15 March 2026)
Weekly Activities	Completed the Project Proposal document.
Individual Contributions	I contributed by providing database-related information for the proposal, including the types of data the system would manage and how the database would support system functions.
Challenges	The challenge was ensuring that the database requirements matched the proposed system features without creating unnecessary complexity.
Action Plans	Begin designing the database schema and prepare an initial ERD for the architecture phase.

Week 6

Item	Description
Week	Week 6 (16 - 22 March 2026)
Weekly Activities	The team started working on the architecture design while database planning continued. The system components and data flow between modules were reviewed.
Individual Contributions	I designed the initial database structure by identifying tables, attributes and relationships. I worked on the relationship between users and disaster reports, as well as how volunteer tasks and notifications would connect to users.
Challenges	Creating relationships between tables was challenging because incorrect relationships could affect the system workflow later during implementation.
Action Plans	Complete the ERD and prepare the database design section for the architecture document.

Week 7

Item	Description
Week	Week 7 (23 - 29 March 2026)
Weekly Activities	The team completed the Architecture Design document including system diagrams, requirements and architecture explanation.
Individual Contributions	I contributed to the database-related sections by explaining how the application would communicate with the database layer. I also reviewed whether the architecture design supported proper data flow between JSP pages, Servlets, services and the database.
Challenges	The main challenge was ensuring that the database structure aligned with the MVC architecture and would support future implementation.
Action Plans	Implement the database structure and prepare the connection between the application and MySQL.

Week 8

Item	Description
Week	Week 8 (30 March - 5 April 2026)
Weekly Activities	Development started and the team began implementing different system modules based on the architecture design.
Individual Contributions	I created the MySQL database structure and assisted in configuring the JDBC connection. I worked on the database connection class and tested whether the application could communicate with the database successfully. I also assisted other members when they needed database-related information for their modules.
Challenges	The main issue faced was configuring the database connection correctly, including checking the JDBC driver, connection URL and database settings.
Action Plans	Continue testing database operations and support backend module development.

Week 9

Item	Description
Week	Week 9 (6 - 12 April 2026)
Weekly Activities	The team prepared for the first project observation by demonstrating the current progress of the system.
Individual Contributions	I prepared the database explanation for the presentation and demonstrated how the application stored and retrieved information.
Challenges	Some features were still under development, making it necessary to verify that the database structure could support future modules.
Action Plans	Improve database testing and continue supporting the implementation of remaining features.

Week 10

Item	Description
Week	Week 10 (13 - 19 April 2026)
Weekly Activities	The team continued developing and integrating the system modules before the prototype submission.
Individual Contributions	I tested database operations involving user records, disaster reports and notifications. I also assisted in checking whether the data displayed in the JSP pages matched the information stored in the database.
Challenges	The challenge was identifying whether errors came from the database, backend logic, or frontend display because the modules were connected.
Action Plans	Perform more integration testing and verify database consistency before prototype submission.

Week 11

Item	Description
Week	Week 11 (20 - 26 April 2026)
Weekly Activities	The team prepared the working prototype submission with screenshots, repository evidence and demonstration materials.
Individual Contributions	I verified that the database functions required for the prototype were working correctly. I checked user login data, report retrieval and other

	database-related workflows before the demonstration. requirements. Assisted team members with final testing before submission.
Challenges	Ensuring all modules could access the correct database information during demonstration was challenging because the system depended on multiple components working together.
Action Plans	Continue database validation and support final integration testing.

Week 12

Item	Description
Week	Week 12 (27 April – 3 May 2026)
Weekly Activities	The team conducted the second project observation and demonstrated additional completed features.
Individual Contributions	I assisted with final database testing and checked data consistency between different modules. I also helped identify issues related to incorrect data retrieval and worked with other members to resolve them.
Challenges	The challenge was fixing integration issues while preparing the final report at the same time.
Action Plans	Complete database documentation and finalize the report content.

Week 13

Item	Description
Week	Week 13 (4 – 10 May 2026)
Weekly Activities	The team completed the final project report and individual logbooks for submission.
Individual Contributions	I documented the database implementation process, including database structure, connectivity and testing results. I also reviewed the final system to ensure stored data and displayed information were consistent.
Challenges	The main challenge was ensuring the final documentation accurately described the implemented database features.
Action Plans	Prepare for the final demonstration and perform final system checks.

3.0 GenAI Usage Record

Week	Tool Used	Purpose	Prompt Used	Output Summary	How It Was Used
Week 3	ChatGPT	Database planning and requirement analysis	"What database entities are commonly needed for a disaster management system?"	Suggested possible entities such as users, disaster reports, volunteers, resources, tasks and notifications.	Used the suggestions as a reference when identifying the initial database entities for the system.
	ChatGPT	Understand database relationships	"How do I identify relationships between tables in a database?"	Explained one-to-one, one-to-many and many-to-many relationships with examples.	Used the explanation when discussing possible relationships between users, reports and tasks.
Week 4	ChatGPT	Database requirement analysis	"What database requirements should be included in a software project proposal?"	Provided examples of data storage, security and data management requirements.	Used the information to assist in preparing the database-related section of the project proposal.
Week 5	ChatGPT	Learn database normalization	"Explain database normalization and why it is important."	Explained requirements such as security, usability, reliability and performance.	Explained normalization concepts including reducing data redundancy and improving consistency.

Week 6	ChatGPT	ERD design support	"How can I create an ERD for a system with users, reports and tasks?"	Suggested entity structures and possible relationships between tables.	Used the output as a reference while designing the initial Entity Relationship Diagram.
	ChatGPT	SQL table design	"What are the important fields for a user management table?"	Suggested fields such as user ID, name, email, password, phone number and role.	Used the suggestions when designing the user table structure in MySQL.
	ChatGPT	Architecture documentation support	"How should database components be explained in an architecture document?"	Explained how databases interact with application layers and backend services.	Used the explanation when writing the database section of the architecture design document.
Week 8	ChatGPT	Testing planning	"What are important test cases for a Java web application?"	Suggested testing areas such as login, validation, navigation and database operations.	Used the troubleshooting steps while testing the DBConnection.java file and resolving connection issues.
Week 9	ChatGPT	JDBC connection troubleshooting	"Why is my Java application unable to connect to MySQL using JDBC?"	Suggested checking driver configuration, connection URL, username and database settings.	Used the structure to organize materials for Project Observation 1.

Week 11	ChatGPT	SQL query improvement	"How can I write SQL queries to retrieve related records from multiple tables?"	Explained the use of SELECT statements, JOIN operations and filtering conditions.	Used the explanation when preparing queries for retrieving reports and user information.
	ChatGPT	Database testing preparation	"What database test cases should be performed for a web application?"	Suggested testing insert, update, delete, retrieval and invalid input scenarios.	Used the suggestions to prepare database testing scenarios before Project Observation 1.
Week 12	ChatGPT	Data validation support	"How can I validate user input before storing data into a database?"	Suggested checking empty values, incorrect formats and invalid data types.	Used the information when reviewing form inputs before database operations.
Week 13	ChatGPT	Prototype database verification	"What should be checked when testing database integration with a prototype?"	Explained checking database connection, data flow and consistency between modules.	Used the checklist during prototype testing before Deliverable 3 submission.
	ChatGPT	Debugging database issues	"How can I troubleshoot incorrect data being retrieved from MySQL?"	Suggested checking SQL queries, conditions, table relationships and stored values.	Used the steps to investigate data retrieval issues during integration testing.

	ChatGPT	Database documentation support	"How to write a database implementation section for a software report?"	Provided guidance on explaining database design, tables and implementation details.	Used the structure as reference when writing the database section in the final report.
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